

Aivory VPS Runtime Documentation

Overview

Aivory runs on a VPS where the frontend sends requests into the ZeroClaw Gateway, which loads the active identity layer and routes workflow-related requests through the workflow skill path. Workflow execution is system-controlled: ZeroClaw handles reasoning and routing, MCP Bridge handles orchestration, and `n8n-as-code-service` handles draft build, sandbox validation, setup report generation, and cleanup.

Active Paths

The active ZeroClaw root is `/home/ubuntu/.zeroclaw`, and the active workspace is `/home/ubuntu/.zeroclaw/workspace`. The shell aliases configured in `.bashrc` are `ZC_ROOT=/home/ubuntu/.zeroclaw`, `ZC_WS=/home/ubuntu/.zeroclaw/workspace`, and `ZC_SKILLS=/home/ubuntu/.zeroclaw/workspace/skills`.

text

```
/home/ubuntu/.zeroclaw/                                # ZC_ROOT
├─ config.toml                                           # active
zeroclaw config
├─ config.toml.backup*                                  # backups
├─ otp-secret
├─ .secret_key
├─ daemon_state.json
└─ workspace/                                           # ZC_WS
    ├─ AGENTS.md                                         # active
operating rules
    └─ IDENTITY.md                                       # active
identity/persona
    └─ SOUL.md                                           # active
behavioral layer
    ├─ HEARTBEAT.md
    ├─ devices.db
    ├─ cron/
    └─ memory/
```

```

├─ sessions/
├─ state/
└─ skills/                                     # ZC_SKILLS
    ├─ multi-agent-router/
    │   ├─ manifest.toml
    │   ├─ README.md
    │   └─ SKILL.md
    ├─ workflow-copilot/
    │   ├─ manifest.toml
    │   ├─ README.md
    │   └─ SKILL.md
    ├─ n8n/
    │   ├─ manifest.toml
    │   ├─ README.md
    │   └─ SKILL.md
    └─ ...

```

The active identity layer files are `/home/ubuntu/.zeroclaw/workspace/AGENTS.md`, `/home/ubuntu/.zeroclaw/workspace/IDENTITY.md`, and `/home/ubuntu/.zeroclaw/workspace/SOUL.md`. The VPS Bridge project path is `/home/ubuntu/AVRY/vps-bridge`, and the systemd service file is `/etc/systemd/system/vps-bridge.service`.

Ports and Services

The ZeroClaw Gateway listens on `127.0.0.1:3010`, and MCP Bridge listens on `127.0.0.1:3020`. Deployment to the user's n8n instance only happens after explicit user approval through the Activate flow, and the user provides the host URL and API key at that stage.

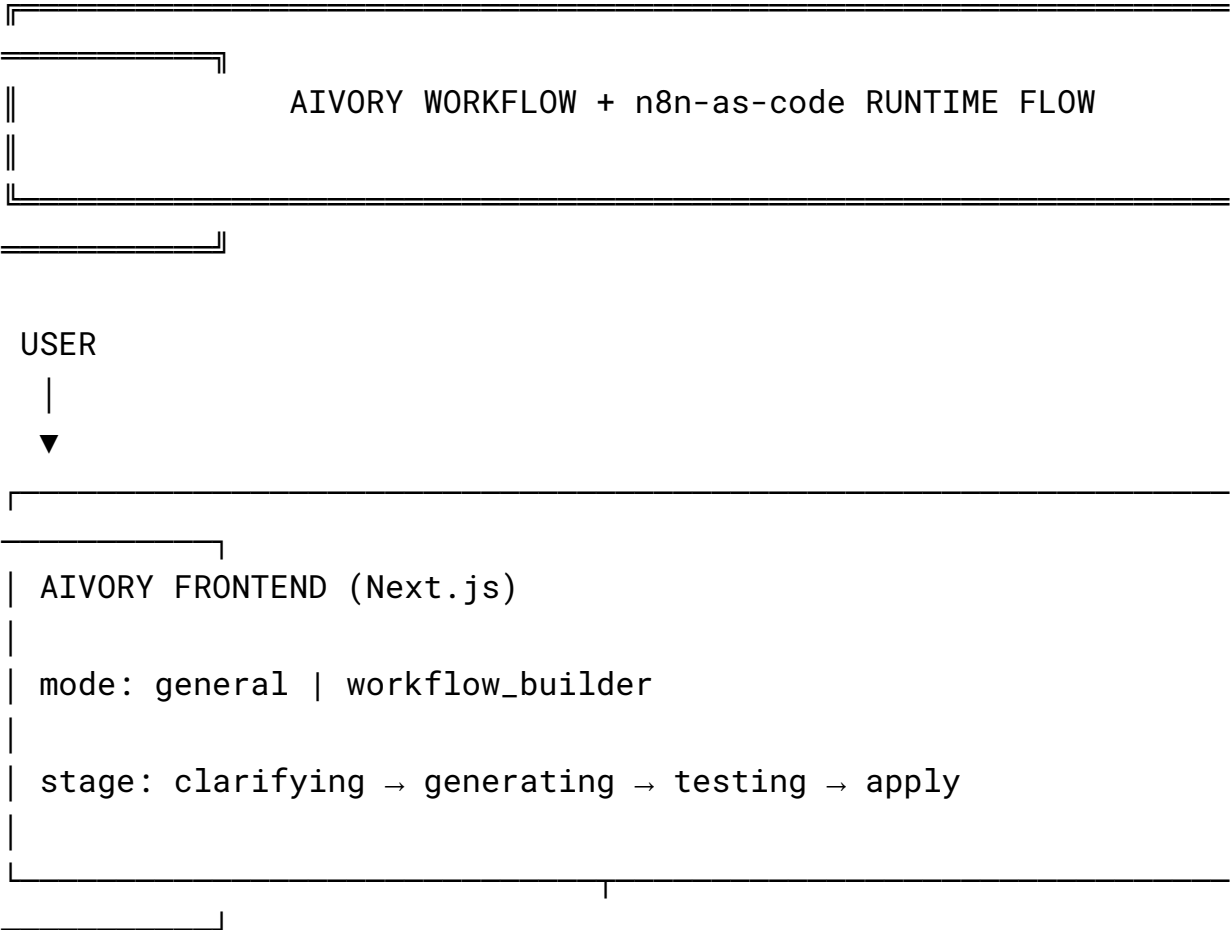
Service	Bind	Purpose
ZeroClaw Gateway	127.0.0.1:3010	Entry point from Aivory frontend into Aivory and Workflow Brain runtime

MCP Bridge	127.0.0.1:3020	Orchestrator, auth token layer, and pass-through to <code>n8n-as-code-service</code>
User n8n Instance	User host	Final deploy target only after Activate approval

Runtime Flow

Workflow requests enter through the frontend, pass into ZeroClaw, load the identity layer, route through the workflow-related skills, and then move into MCP Bridge for execution orchestration. The `n8n-as-code-service` performs build draft, sandbox validation, setup report generation, and cleanup before any final deployment is allowed.

text



| HTTP / WebSocket



| ZEROCLAW GATEWAY

| 127.0.0.1:3010

| config: /home/ubuntu/.zeroclaw/config.toml

| workspace: /home/ubuntu/.zeroclaw/workspace/

| loads identity layer



| IDENTITY LAYER

| /home/ubuntu/.zeroclaw/workspace/IDENTITY.md

| /home/ubuntu/.zeroclaw/workspace/SOUL.md

| /home/ubuntu/.zeroclaw/workspace/AGENTS.md

| route by intent



| SKILLS

| /home/ubuntu/.zeroclaw/workspace/skills/multi-agent-router/

| /home/ubuntu/.zeroclaw/workspace/skills/workflow-copilot/

| /home/ubuntu/.zeroclaw/workspace/skills/n8n/

| workflow / n8n request



| MCP BRIDGE

| 127.0.0.1:3020

| transport: HTTP | auth: token



| n8n-as-code-service

| build draft → sandbox validate → setup report → cleanup

| only after user clicks Activate



USER'S SELF-HOSTED n8n INSTANCE
host URL + API key user

Key Files

The workflow path depends primarily on the `multi-agent-router`, `workflow-copilot`, and `n8n` skills stored in the active workspace skills directory. The `AGENTS`, `IDENTITY`, and `SOUL` files form the active agent identity and behavior layer injected during runtime startup.

File	Path	Role
Active config	<code>/home/ubuntu/.zeroclaw/config.toml</code>	Main ZeroClaw runtime configuration
AGENTS	<code>/home/ubuntu/.zeroclaw/workspace/AGENTS.md</code>	Operating rules and hard guardrails
IDENTITY	<code>/home/ubuntu/.zeroclaw/workspace/IDENTITY.md</code>	Persona and identity rules
SOUL	<code>/home/ubuntu/.zeroclaw/workspace/SOUL.md</code>	Behavioral and workflow architecture layer

Multi-agent router skill	/home/ubuntu/.zeroclaw/workspace/skills/multi-agent-router/SKILL.md	Intent-based routing into specialist path
Workflow Copilot skill	/home/ubuntu/.zeroclaw/workspace/skills/workflow-copilot/SKILL.md	Workflow orchestration and stage flow
n8n skill	/home/ubuntu/.zeroclaw/workspace/skills/n8n/SKILL.md	n8n-oriented generation path
Bridge service file	/etc/systemd/system/vps-bridge.service	systemd unit for VPS Bridge
Bridge project	/home/ubuntu/AVRY/vps-bridge	Bridge implementation directory

Operations

The safest operational pattern is to verify aliases, config, identity files, workflow skills, listening ports, and bridge status before changing any workflow-related logic. The commands below are focused on inspection and are suitable for routine runtime checks.

```
bash
```

```
# Aliases
```

```
echo $ZC_ROOT
```

```
echo $ZC_WS
```

```
echo $ZC_SKILLS
```

```
# Active config
```

```
ls -lah /home/ubuntu/.zeroclaw/
```

```
sed -n '1,220p' /home/ubuntu/.zeroclaw/config.toml
```

```
# Identity layer
```

```
ls -lah /home/ubuntu/.zeroclaw/workspace/
```

```
sed -n '1,220p' /home/ubuntu/.zeroclaw/workspace/AGENTS.md
```

```
sed -n '1,220p' /home/ubuntu/.zeroclaw/workspace/IDENTITY.md
```

```
sed -n '1,220p' /home/ubuntu/.zeroclaw/workspace/SOUL.md
```

```
# Workflow skills
```

```
find /home/ubuntu/.zeroclaw/workspace/skills -maxdepth 2 -type f  
| sort
```

```
sed -n '1,220p'
```

```
/home/ubuntu/.zeroclaw/workspace/skills/workflow-copilot/SKILL.m  
d
```

```
sed -n '1,220p'
```

```
/home/ubuntu/.zeroclaw/workspace/skills/n8n/SKILL.md
```

```
sed -n '1,220p'
```

```
/home/ubuntu/.zeroclaw/workspace/skills/multi-agent-router/SKILL  
.md
```

```
# Open ports
```

```
ss -lntp | grep -E '3010|3020'
```

```
# Bridge service
```

```
systemctl status vps-bridge --no-pager
```

```
sed -n '1,220p' /etc/systemd/system/vps-bridge.service
```

```
ls -lah /home/ubuntu/AVRY/vps-bridge
```

Troubleshooting

If workflow generation does not behave correctly, the first checks should be whether the active `config.toml` is the correct one, whether `AGENTS.md`, `IDENTITY.md`, and `SOUL.md` are present in the active workspace, and whether the workflow-related skills exist under the active `skills/` directory. If execution breaks after routing, the next checks should be whether ZeroClaw is bound on `3010`, MCP Bridge is bound on `3020`, and whether the bridge service is healthy under `systemd`.

A deployment issue after successful draft generation usually points to the Activate path or downstream user n8n connectivity, because actual deploy only occurs after explicit Activate approval and uses the user's own host URL and API key. A generation issue before that point is more likely to be in the skill-routing, identity-layer, or bridge-orchestration path.